Price Elasticity Across Weather Conditions and Time Slots

A simple explanation of the two professor-discussion documents

1. Document 1 — Mean, standard deviation, and clustering

The first recording explains that adding more data can increase the variance (spread) of elasticity values. The professor suggests calculating the mean and standard deviation of price elasticity separately for each weather scenario, then using clustering to identify scenarios with similar elasticity. Scenarios may have different numbers of rows, so keep their sample counts visible.

Step 1: Analyze each scenario separately

For example, calculate elasticity separately for clear, cloudy, and rainy weather. Do not assume customers react to prices in the same way under every condition. These are example scenario categories; the actual results must come from your dataset.

Step 2: Calculate the mean and standard deviation

Mean elasticity is the average elasticity in a scenario:

Mean elasticity = (sum of all elasticity observations) / n

Here, n is the number of valid elasticity observations. The mean summarizes the average price sensitivity.

Standard deviation describes how widely the elasticity values are spread around their mean. A smaller value indicates values are more closely grouped; a larger value indicates greater spread.

Weather	Illustrative mean elasticity	Illustrative standard deviation
Clear	-0.8	0.5
Cloudy	-0.9	0.6
Rainy	-1.8	0.7

Note: These numbers are illustrative only and are not results from your dataset.

Step 3: Cluster similar scenarios

Clustering means grouping scenarios whose elasticity statistics are similar. For example, clear and cloudy weather might be candidates for one group if their mean elasticity and spread are close. Rainy weather might remain separate if its elasticity differs. The grouping must be supported by a defined similarity measure, not just by the weather labels.

2. Document 2 — How to combine scenarios correctly

The second document says to calculate mean and standard deviation for each weather condition or time slot separately, then combine scenarios based on a similarity index or clustering. The professor explicitly says not to combine the raw data and train them together. This analysis is for understanding customer sensitivity.

Step 1: Calculate statistics for each scenario

For every scenario (for example, clear-morning, clear-evening, cloudy-morning, or rainy-evening), record the observation count, mean elasticity, and standard deviation.

Step 2: Compare scenarios

A distance measure can help compare scenarios. One possible example is Euclidean distance using mean elasticity and standard deviation:

$d(A,B) = \text{sqrt}[(\text{MeanA} - \text{MeanB})^2 + (\text{StdA} - \text{StdB})^2]$

A smaller distance indicates greater similarity. The documents do not prescribe a specific clustering algorithm or similarity threshold, so that choice must be defined and justified in the implementation.

Step 3: Check the combined standard deviation

The professor's stated rule is that a combined scenario is acceptable only when its standard deviation is strictly below 4. If it is 4 or greater, exclude that combination.

Combined standard deviation	Decision
2.6	Accept — below 4
4.0	Exclude — the limit is strictly below 4
4.3	Exclude — above 4

Technical note: calculate the standard deviation of the combined elasticity observations, or use an equivalent pooled calculation. Do not simply average the individual scenario standard deviations.

3. What not to do

Correct: calculate elasticity statistics separately, compare scenarios, and combine similar scenarios for analysis only after checking the standard-deviation threshold.

Incorrect: merge all raw records from different weather scenarios and train them together as if the conditions were interchangeable. That could hide differences in customer sensitivity.

4. Recommended implementation procedure

1. Clean the complete dataset

Check missing or invalid values and ensure fare, demand, weather, and time information are usable.

2. Define scenarios

Separate observations by the selected weather classes and/or time slots.

3. Calculate elasticity

Apply the approved price-elasticity formula separately to each scenario.

4. Calculate statistics

Record observation count, mean elasticity, and standard deviation for every scenario.

5. Compare scenarios

Use a defined similarity index or clustering method to identify candidate groups.

6. Test combinations

Calculate combined elasticity statistics without merging raw records for joint model training.

7. Apply the threshold

Keep a combination only if its standard deviation is below 4; otherwise exclude it.

8. Report findings

Present individual results, accepted and excluded groups, and the implications for customer sensitivity.

5. Suggested scenario-level results table

Column	Purpose
Scenario_ID	Unique scenario identifier
Weather_Condition	Clear, cloudy, rainy, etc.
Time_Slot	Defined time period
Observation_Count	Number of valid elasticity observations
Mean_Elasticity	Average price sensitivity
Std_Elasticity	Spread of elasticity values
Cluster_ID	Cluster assigned by the selected method
Combined_Scenarios	Scenarios in a candidate combination
Combined_Std	Standard deviation after combination
Decision	Accept if below 4; otherwise exclude

These are suggested implementation columns, not a list explicitly required by the professor.

6. Difference between the two documents

Document 1	Document 2
Explains why adding data may increase variability.	Explains how to combine similar scenarios and apply the threshold.
Recommends comparing mean and standard deviation.	Recommends similarity-based grouping or clustering.
Notes that scenarios may have different row counts.	Says not to combine raw data for joint training.

Final takeaway

The research goal is to understand how customer price sensitivity changes across weather conditions and time slots. First analyze each scenario separately, identify similar scenarios, and then test candidate combinations against the standard-deviation limit of less than 4. DQN, policy networks, and reinforcement learning are not part of this specific analysis step; complete and validate the elasticity analysis first.